Pratham Sharma

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SKILLS

- Languages: Java, C++, JavaScript, Python
- Frameworks: CSS, Node.js, React.js, HTML
- Tools/Platforms: MySQL, Anaconda, MongoDB, Git, GitHub, Google Colab
- Soft Skills: Adaptability, Teamwork, Problem-Solving, Quick Learner

PROJECTS

- **Smart Product Recommendation System for E-commerce | GitHub** November 2025 - Present
- Designed an intelligent recommendation engine to personalize product suggestions based on user activity and purchase behaviour.
- Engineered a collaborative filtering model and connected it to a dynamic frontend via Flask REST APIs.
- Elevated model precision by 12% through data refinement and advanced parameter optimization.
- Tech Stack: Python, Pandas, NumPy, Scikit-Learn, Flask, Matplotlib, Google Colab.

- **Deadlock Detector and Resolver Tool: | GitHub** May 2025

- Built and implemented a Python-based tool for the detection and resolution of resource contention in concurrent systems.
- Utilized threading, synchronization, and concurrency to manage resources efficiently.
- Improved reliability by 40%, increased throughput by 30%, and eliminated 95% of deadlock occurrences through automated detection and resolution.
- Tech Stack: Python, Threading, Concurrency, Synchronization, Data Structures.

- **HealthCare – ML-based Disease Prediction System | GitHub** December 2024

- Architected a Machine Learning based disease risk prediction platform with structured data pipelines.
- Optimized datasets through comprehensive preprocessing and feature engineering in Python.
- Achieved accurate and reliable prediction of chronic disease risks, supporting early detection and data-driven healthcare decision-making.
- Tech Stack: Python (Pandas, NumPy, Scikit-learn), HTML, CSS, MySQL.

- **Water Pollution Monitoring & Awareness Website | GitHub** January 2024
- Deployed a web-based platform to analyse and visualize water pollution data, highlighting contamination levels and pollution sources across regions.
- Integrated data processing and validation features to ensure accurate representation of water quality parameters such as pH, turbidity, and dissolved oxygen.
- Enabled clear identification of high-risk water pollution zones through interactive visual insights, supporting data-driven environmental awareness and decision-making.
- Tech Stack: HTML, CSS, JavaScript, Python (Flask), Data Visualization, Environmental Data.

CERTIFICATES

- ChatGPT-4 Prompt Engineering | ChatGPT, Generative AI & LLM | (Certificate) September 2025
- Foundations of Data Structures in C, C++ and Java | (Certificate) July 2025
- Java Programming | Iamneo |(Certificate) May 2025
- Object Oriented Programming | iamneo | (Certificate) December 2024
- Computer Communications | Coursera | (Certificate) November 2024
- The Complete Full-Stack Web Development Bootcamp | Udemy | (Certificate) February 2024

ACHIEVEMENTS

- Solved 150+ algorithmic problems on LeetCode: August 2025
  Strengthening Problem-Solving skills and understanding of Data Structures and Algorithms.
- Solved 250+ algorithmic problems on NeoColab: December 2024
  Demonstrating effective Problem-Solving skills in Java challenges.

EDUCATION

Lovely Professional University Punjab, India
Bachelor of Technology - Computer Science and Engineering; CGPA: 5.9 August 2023 - Present

Little Flower Senior Secondary School Kangra, HP
Intermediate; Percentage: 74 April 2021 - March 2022

Little Flower Senior Secondary School Kangra, HP
Matriculation; Percentage: 84 April 2019 - March 2020